

Inside Winnower: Component Semantics of Cellular-Automaton Background Decomposition, and the Exact Boundary of Period Inflation

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Abstract

A recurring task in artificial life is separating a life-like pattern from the periodic domain it lives in: a glider is a glider only relative to its background. *Winnower* is an open-source tool that performs this split on a cellular-automaton (CA) spacetime, fitting a drifting periodic *background* by majority vote and returning the *residual mask* of cells the background cannot explain. We give a component view of the tool in which each stage of the pipeline — candidate, orbit classes, majority vote, background, residual mask, and a complexity-aware score — is a diagram element carrying an explicit semantic role. We then address a pressure common to every fit-based detector, *period inflation*: a longer-period candidate has more parameters and can never fit worse, so a selector scoring by fit alone drifts to the largest model. We report a result that pins down exactly when this pressure exists. For two constant-velocity candidates, five natural conditions — velocity-matched divisibility, orbit-partition refinement, model-class nesting, universal residual improvement, and universal likelihood improvement — are all equivalent, and divisibility is *necessary*, not merely sufficient: whenever it fails, some spacetime makes the longer candidate strictly worse. The result is proved directly and confirmed by an exhaustive check over all candidate pairs on a small grid. It is the precise justification for Winnower’s complexity penalty, and it is reproducible in the tool’s live web companion.

Introduction

Cellular automata routinely settle into broad periodic domains threaded with defects, interfaces, and particle-like activity. For artificial life those domains are not noise to be discarded but the *reference frame* that makes coherent structure legible: gliders, spaceships, and their collisions are defined against the background they traverse (Hanson and Crutchfield, 1992; Crutchfield and Hanson, 1997). Extracting that background at scale is a prerequisite for cataloguing where life-like behaviour appears — a concern squarely within the ALIFE 2026 theme of living and life-like complex adaptive systems.

Winnower is a fast global front end for this problem. Given a binary spacetime U and a grid of *candidates* (p, s) — “the pattern repeats every p time steps, drifting sideways by s ” — it groups the cells each candidate forces to

be equal (its *orbit classes*), takes a majority vote in each class to build the best-fitting background B^* , and records the residual mask $M = U \oplus B^*$. Each candidate costs one linear pass, so whole-catalogue 1D/2D/3D surveys are practical, and every reported number re-runs in a browser-based reproduction; the pipeline and full experimental evaluation are in the companion paper (Herman, 2026). This abstract gives a component view assigning each stage its semantic role (fig. 1) and reports an exact characterization of when the selector’s complexity penalty is needed.

The Pipeline, Stage by Stage

Figure 1 names each stage of Winnower and the meaning it carries. A *candidate* (p, s) is read as a spacetime translation τ ; its *orbit classes* Π are the cells τ forces to be equal; a per-class *majority vote* is the Hamming-optimal projection of U onto the model class $\mathcal{B}(p, s)$ of fields constant on classes, yielding the *background* B^* and the *residual mask* $M = U \oplus B^*$; and each candidate is scored by a description length $\text{NML} = \text{NLL} + \text{complexity}$, the Bernoulli fit plus a normalized-maximum-likelihood penalty for capacity (Shtarkov, 1987; Grünwald, 2007), with *selection* returning the lowest-scoring candidate and its margin. Because every box corresponds to a named object, the figure doubles as the legend for the rest of this abstract.

An Exact Boundary for the Penalty

The one design choice that needs justifying is the complexity penalty. Without it the score reduces to fit, and fit alone is biased. Call (p_2, s_2) a *velocity-matched multiple* of (p_1, s_1) when $p_2 = m p_1$ and $s_2 = m s_1 \bmod \mathbf{D}$: the same drift at a longer period. Such a multiple splits every orbit class into finer pieces, so its optimal residual count and its NLL can only improve, and an unpenalized selector inflates the period whether or not extra structure is present. We show exactly when this pressure exists.

Fix a torus with spatial dimensions $\mathbf{D}$ and temporal length T , and two candidates c_1, c_2 acting by translation. Write Π_i for the orbit partition of c_i , $\mathcal{B}_i$ for the model class of fields constant on Π_i -classes, $d_U^*(c_i)$ for the minimum

Relative-periodic domain selection by Bernoulli NML

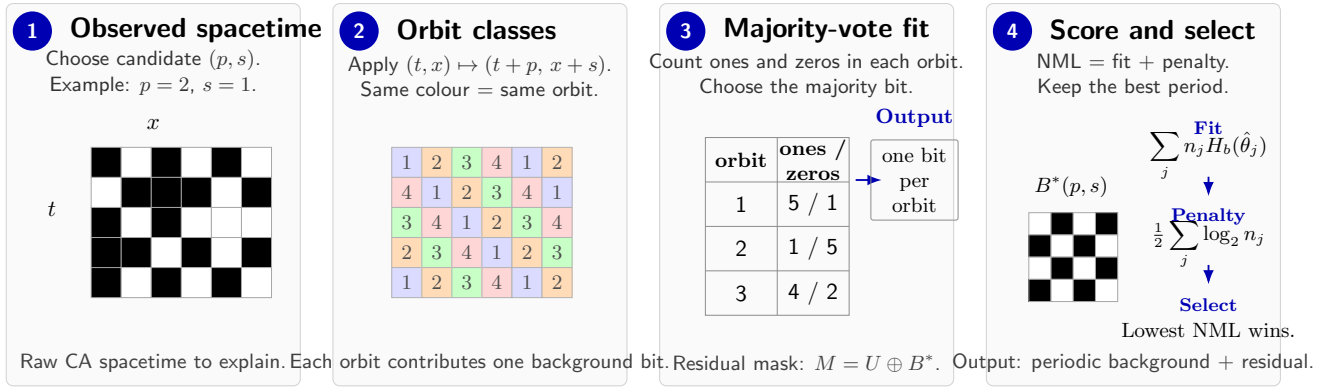


Figure 1: How Winnower decomposes a spacetime, in four stages — each panel a stage of the pipeline paired with the object it computes and its semantic role. **(1)** A candidate (p, s) is chosen and applied to the observed spacetime U . **(2)** The induced translation $(t, x) \mapsto (t + p, x + s)$ groups cells into orbit classes (same colour = same orbit); each class will contribute one background bit. **(3)** A per-class majority vote fixes that bit, giving the background B^* and the residual mask $M = U \oplus B^*$ — the cells the background cannot explain. **(4)** Each candidate is scored by a Bernoulli NML description length — fit $\sum_j n_j H_b(\hat{\theta}_j)$ plus a complexity penalty $\frac{1}{2} \sum_j \log_2 n_j$ — and the lowest-scoring period is selected, returning the periodic background and its residual. Every symbol ($U, (p, s), B^*, M$) names an entity referred to in the text, so the figure doubles as a legend; full derivations are in the companion paper (Herman, 2026).

Hamming distance from U to B_i , and $\text{NLL}_U(c_i)$ for the orbit-class Bernoulli negative log-likelihood. The following are equivalent: (i) c_2 is a velocity-matched multiple of c_1 ; (ii) c_2 's translation is a power of c_1 's; (iii) Π_2 refines Π_1 ; (iv) $B_1 \subseteq B_2$; (v) $d_U^*(c_2) \leq d_U^*(c_1)$ for every U ; (vi) $\text{NLL}_U(c_2) \leq \text{NLL}_U(c_1)$ for every U .

The forward chain (i) $\Rightarrow \dots \Rightarrow$ (vi) is routine (cyclic-group structure; Jensen's inequality for the concavity of binary entropy). The content is *necessity*: if divisibility fails, refinement fails, and a concrete adversarial spacetime — constant on Π_1 -classes but mixing two of them inside one Π_2 -class — makes c_2 strictly worse under both d^* and NLL. Closing the loop needs (iii) $\Rightarrow$ (ii), that refinement *forces* a power: on the torus, orbits are cosets of the cyclic subgroup a translation generates, so refinement at the identity gives subgroup containment, and a translation is fixed by its value at one point. Divisibility is thus the exact frontier of the guaranteed-improvement regime — on one side longer periods always win, on the other the data can always punish them — and hence exactly the set of comparisons the penalty of fig. 1's Score stage must neutralize. We verify the finite-window form of the equivalence by exhaustively enumerating every candidate pair on a 6×4 space-time grid.

Reproducible in the Tool

The characterization is visible where the tool lives. The same in-browser reproduction that backs every reported number displays, for any supported rule, seed, and hori-

zon, the per-period fit (NLL) and penalized (NML) scores side by side, so a visitor can watch the fit column improve monotonically along velocity-matched chains — exactly as predicted — while the penalized column resists. At survey scale the effect is stark: on the 105 non-degenerate Life-like rules at horizon $T=100$, selecting by fit alone picks the largest scanned period for *every* rule, whereas the penalized score concentrates selections at low period.

Status and Future Work

This is work in progress. The empirical picture — planted-ground-truth recovery at 1.00 joint accuracy over 642 runs, zero false positives on shuffled and random controls — is established in the companion paper (Herman, 2026). Open threads include extending the analysis from pairwise comparisons to the full candidate lattice the scanner searches, and relaxing the nondegeneracy hypothesis ($p_2 \leq T - 1$) in the finite-window form. The tool, code, tests, and in-browser reproduction are public.

References

- Crutchfield, J. P. and Hanson, J. E. (1997). Computational mechanics of cellular automata: An example. *Physica D*, 103:169–189.
- Grünwald, P. (2007). *The Minimum Description Length Principle*. MIT Press.
- Hanson, J. E. and Crutchfield, J. P. (1992). The attractor-basin portrait of a cellular automaton. *Journal of Statistical Physics*, 66:1415–1462.

Herman, J. (2026). Winnower: A fast tool for splitting cellular automaton spacetimes into a periodic background and a residual mask. Companion manuscript; source, data, and in-browser reproduction at <https://github.com/zitterbewegung/Winnower>.

Shtarkov, Y. (1987). Universal sequential coding of single messages. *Problems of Information Transmission*, 23:3–17.